BIRZEIT UNIVERSITY

Faculty of Engineering and Information Technology

Data Science Course

Phase 3 Project Report

Analyzing Flight Delays Using Weather Data

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1) Introduction

Flight delays are among the most common problems in the air transport sector, directly impacting airlines and passengers and causing operational disruptions, lost time, and increased costs. These delays are linked to a number of different factors, most notably weather conditions, airport congestion, air traffic, and other operational challenges. Analyzing flight data helps to better understand these factors and uncover patterns associated with delays.

In this project, we relied on a massive dataset of flight delays, combined with weather data, to study the impact of weather on flight performance. The original dataset comprised approximately six million records, making direct analysis computationally challenging. Therefore, the data was first carefully collected and examined, then cleaned, removing unnecessary records. After this stage, a random sample of 25,000 records was selected from the cleaned data for use in subsequent analysis phases.

The project included an exploratory data analysis, the extraction of summaries and descriptive statistics for various variables, and the creation of a set of graphs and visualizations that helped to better understand the nature and distribution of the data. The potential relationships between different weather variables and flight delays were also investigated to identify the factors most strongly associated with delays and to uncover significant patterns within the data.

In the final stage, a predictive model was developed to classify the probability of flight delays based on a selection of weather-related and operational characteristics. This model was used to assess the predictability of delays and to leverage the available data to support decision-making.

This report describes the dataset used, the preprocessing steps applied to it, the analytical and graphical visualization techniques employed throughout the project, and the key findings and conclusions drawn from the study.

2) Dataset Description

This project utilized a dataset obtained from the Kaggle platform, containing detailed flight records along with associated weather data. The original dataset comprised approximately six million records, in addition to a large number of variables describing operational and environmental factors that could affect flight performance and the likelihood of delays.

Due to the large dataset size, several preprocessing steps were necessary before analysis could begin. These steps included verifying data quality, addressing missing values, and cleaning the data of unnecessary or unsuitable records. After the cleaning and preparation phase, a random sample of 25,000 records was selected from the original dataset to facilitate analysis, reduce computational costs, and maintain adequate representation of the underlying data characteristics.

The dataset contains a wide range of variables related to flight performance and factors influencing delays. Key among these variables are data related to weather conditions, aircraft characteristics, airport activity levels, and flight-specific operational information. The analysis focused on several key characteristics, including precipitation amount (PRCP), snowfall rate (SNOW), snow depth (SNWD), maximum temperature (TMAX), average wind speed (AWND), aircraft age (PLANE_AGE), and concurrent flights (CONCURRENT_FLIGHTS).

The target variable in the study was DEP_DEL15, which was used to determine whether a flight experienced a departure delay exceeding 15 minutes. This variable served as the basis for building and analyzing the project's predictive model.

3) Data Cleaning and Preprocessing

3.1 Data Inspection

We initially loaded the dataset into Python using the Pandas library. A preliminary inspection was then performed using functions such as `head()`, `shape()`, and `info()` to understand the dataset structure, identify available attributes, and check the data types.

3.2 Attribute Selection

The original dataset contained a large number of attributes. Therefore, we selected only the variables relevant to flight delays and weather conditions for analysis. These variables included weather measurements such as precipitation (PRCP), snowfall (SNOW), snow depth (SNWD), maximum temperature (TMAX), and average wind speed (AWND), as well as operational variables such as aircraft age (PLANE_AGE), concurrent flights (CONCURRENT_FLIGHTS), and the target variable DEP_DEL15.

3.3 Handling Missing Values

We identified and checked for missing values using the `isnull().sum()` function. We then deleted records containing missing values for critical attributes to ensure data quality and improve the reliability of the analysis results.

3.4 Removing Duplicate Records

We scanned the dataset for duplicate data using the `duplicate()` function. We then removed any duplicate rows found in the dataset to prevent duplicate records from impacting the analysis.

3.5 Sampling the Data

The original dataset contained approximately 6 million flight records. Because it was so large, a subset of approximately 500,000 records was selected and stored for processing. After the cleaning phase, a random sample of 25,000 records was extracted from the dataset. This approach reduced computational requirements while maintaining a representative sample of the original data.

4. Data Exploration and Statistical Summary

After completing the data preprocessing stage, an Exploratory Data Analysis (EDA) was carried out to better understand the dataset and examine its main characteristics before building the predictive model. This step helped provide an initial overview of the data and allowed for the identification of possible patterns and relationships among the different variables.

First, the describe() function was used to generate descriptive statistics for the numerical variables. These statistics included the number of observations, mean, standard deviation, minimum and maximum values, as well as quartile values. This information provided a general understanding of the data distribution and the variability of the different features.

The info() function was also used to review the structure of the dataset and verify the data types of all variables. In addition, the dataset was checked again after the cleaning process to ensure that no missing values or duplicate records remained that could affect the analysis results.

Overall, this stage provided a clearer understanding of the weather-related variables and flight characteristics included in the dataset. It also helped explore their possible relationship with flight delays. The insights obtained from the exploratory analysis served as a foundation for the next stages of the project, including data visualization and predictive modeling.

5. Data Visualization and Interpretation

Data visualization techniques were used to better understand the dataset and explore the relationships between different variables. These visualizations helped present the data in a clearer way, making it easier to identify patterns, trends, and important observations related to flight delays.

Several types of charts and plots were created to examine the distribution of flight delays, analyze weather-related factors, detect outliers, and investigate the relationships among different variables. The visualizations provided useful insights into the characteristics of the data and helped highlight factors that may influence flight delays. The findings obtained from these visual analyses also supported the later stages of the project, including data analysis and predictive modeling.

5.1 Correlation Heatmap

Figure 1 shows the correlation heatmap for the numerical variables included in the dataset. The heatmap was used to examine the relationships between different variables and to see how strongly they are related to each other, especially the weather-related variables, flight characteristics, and the target variable representing flight delays.

From the heatmap, it can be seen that most variables have weak correlations with one another. This suggests that flight delays are not caused by a single factor, but are influenced by several different factors working together. Some moderate relationships can also be observed between weather variables, such as the correlation between snowfall (SNOW) and snow depth (SNWD), which is expected since both variables describe winter weather conditions.

Overall, the correlation heatmap provided a useful overview of the relationships within the dataset. It also helped identify variables that may have an effect on flight delays, making it an important step before moving on to the visualization and predictive modeling stages.

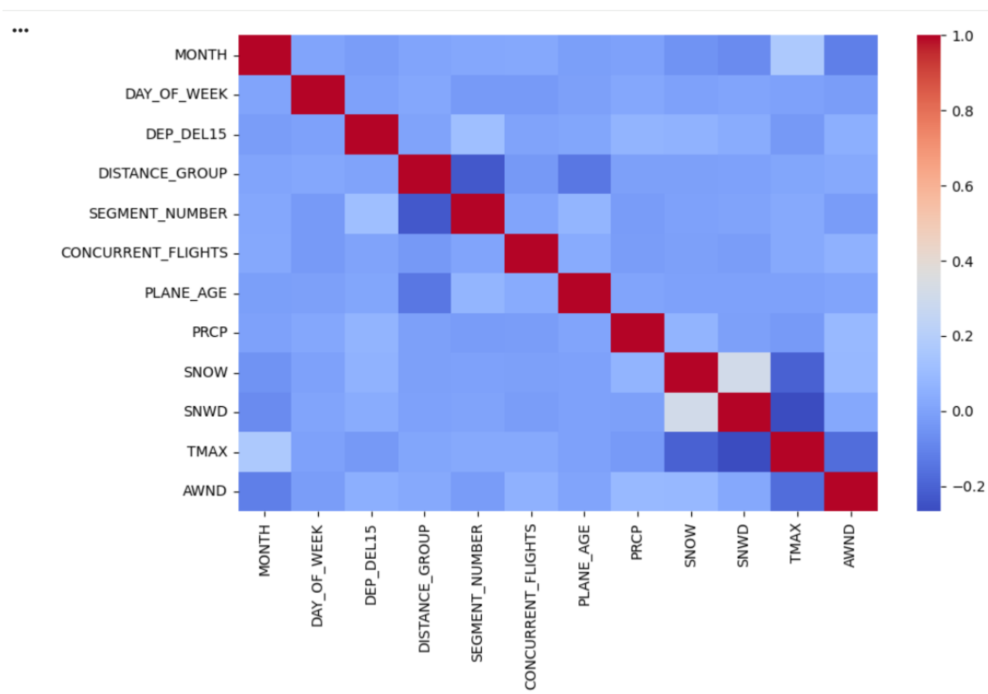

Figure 1Correlation Heatmap

5.2 Distribution of Wind Speed

Figure 2 presents the distribution of wind speed values in the dataset using a histogram. This visualization was created to examine how wind speed measurements are distributed across the flight records and to identify the most common ranges of wind speed.

As shown in the figure, most wind speed values fall within a moderate range, with the highest concentration of observations occurring between approximately 4 and 12 units. The number of records gradually decreases as wind speed increases, indicating that very high wind speeds were much less common than moderate ones.

These results suggest that the majority of flights in the dataset operated under relatively normal weather conditions, while strong wind conditions occurred less frequently. Examining the distribution of wind speed is useful because it provides a better understanding of the weather conditions associated with the flights and helps evaluate whether wind speed may have an impact on flight delays.

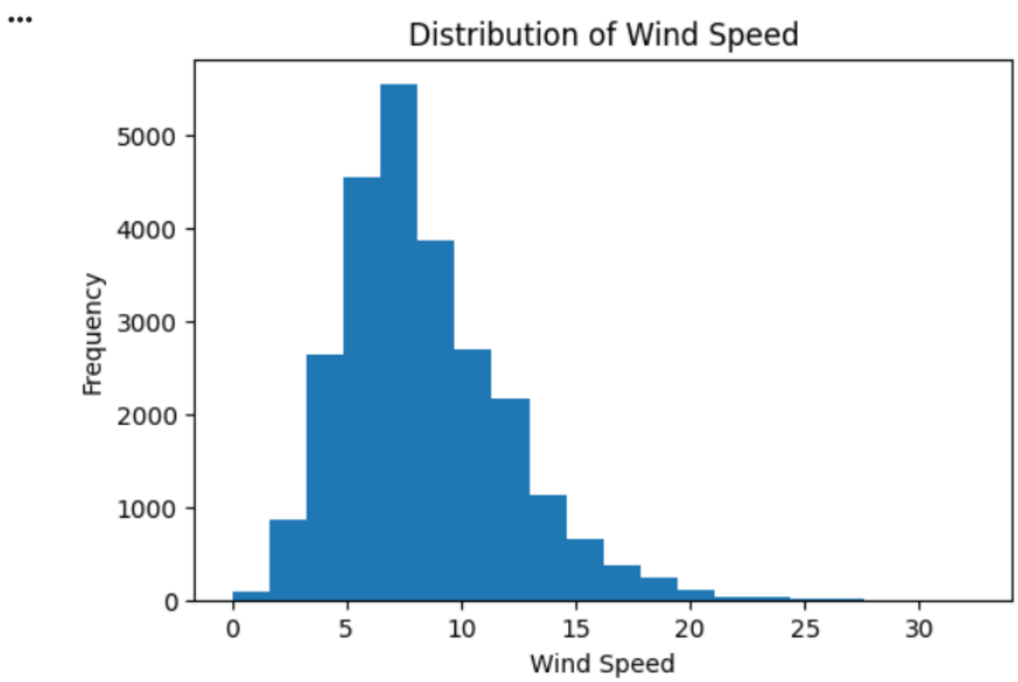

Figure 2: Distribution of Wind Speed (AWND)

5.3 Boxplot of Wind Speed

Figure 3 shows a box plot for the wind speed variable (AWND). This plot was used to gain a clearer understanding of how the values are distributed, how much variation exists in the data, and whether any unusual observations are present.

As shown in the figure, the median wind speed is close to 8 units, and most of the observations are concentrated within a relatively limited range around this value. The plot also highlights several observations that lie above the typical range of the data, indicating the presence of some unusually high wind speed values.

These higher values may represent uncommon weather conditions that occurred during a small number of flights. Overall, the box plot supports the findings from the histogram by showing that most flights were associated with moderate wind speeds, while higher wind speed conditions appeared less frequently in the dataset.

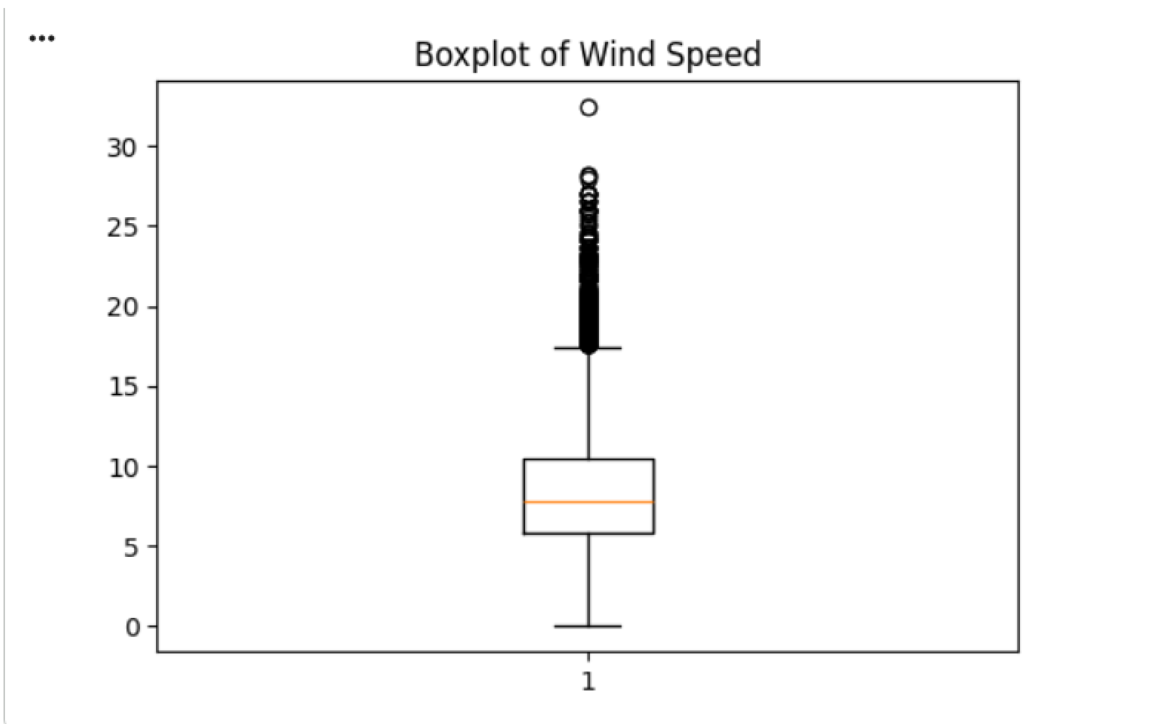

Figure 3: Boxplot of Wind Speed (AWND)

5.4 Relationship Between Wind Speed and Flight Delay

Figure 4 illustrates the relationship between wind speed (AWND) and flight delay status (DEP_DEL15) using a scatter plot. The purpose of this visualization is to explore whether wind speed has any noticeable effect on the likelihood of flight delays.

The target variable (DEP_DEL15) has two possible values: 0 indicates that the flight was not delayed, while 1 indicates that the flight experienced a departure delay of more than 15 minutes. As shown in the figure, both delayed and non-delayed flights appear across different wind speed values.

The scatter plot does not show a clear pattern or a strong relationship between wind speed and flight delays. Delayed flights can be observed at both low and high wind speed levels, suggesting that wind speed alone is not enough to explain why a flight is delayed.

Overall, the results indicate that flight delays are likely influenced by several factors rather than a single variable. While weather conditions may play a role, other operational and environmental factors should also be considered when analyzing or predicting flight delays.

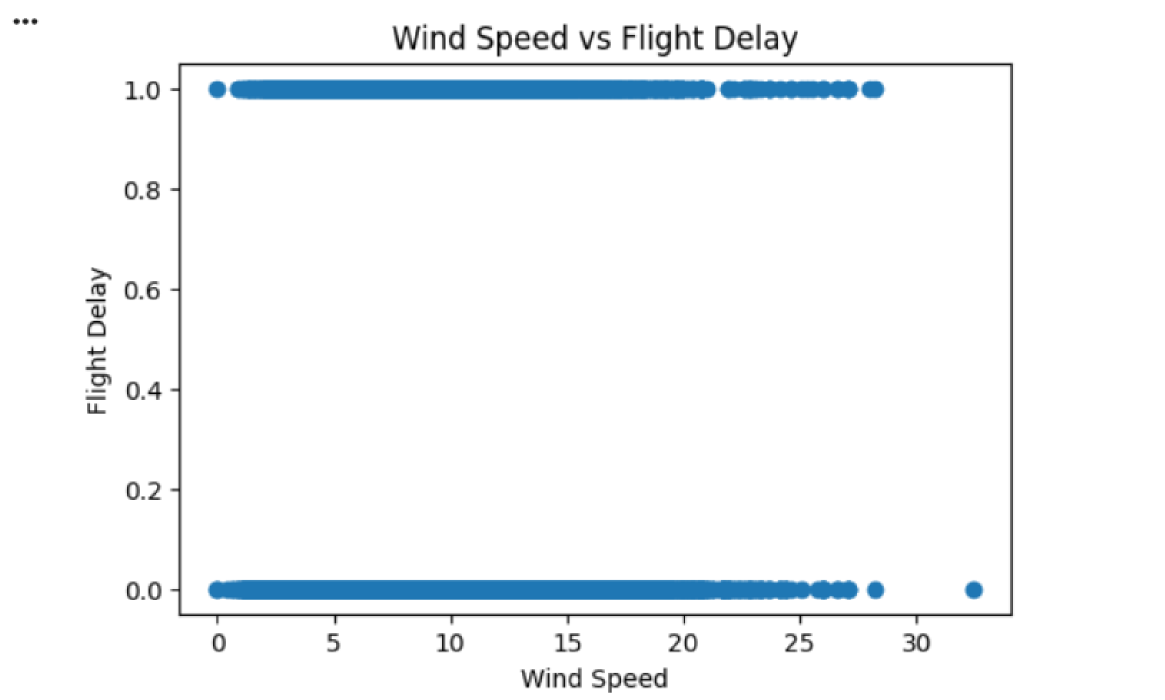

Figure 4 : Relationship Between Wind Speed (AWND) and Flight Delay (DEP_DEL15)

6. Predictive Modeling

To predict flight delays, a Decision Tree Classifier was developed using the Scikit-learn library. The main objective of the model was to determine whether a flight would experience a departure delay of more than 15 minutes based on a set of weather and operational factors.

The model used several input features, including average wind speed (AWND), precipitation (PRCP), maximum temperature (TMAX), aircraft age (PLANE_AGE), and the number of concurrent flights (CONCURRENT_FLIGHTS). The target variable was DEP_DEL15, which indicates whether a flight was delayed by more than 15 minutes at departure.

Before training the model, the dataset was divided into two parts: 80% for training and 20% for testing. The Decision Tree model was then trained using the training data and evaluated on the testing data to measure its performance.

The model achieved an accuracy of 69.4%, meaning it was able to correctly classify a considerable number of flight delay cases. This result suggests that the selected features have a noticeable impact on flight delays, although they do not fully explain the problem. Other factors that were not included in the dataset may also contribute to flight delays.

Overall, the results demonstrate that machine learning techniques can be useful for analyzing and predicting flight delays. They also show that combining weather information with operational flight data can provide valuable insights and support the development of predictive models for delay classification.

7. Conclusion

In this project, a large flight delay dataset combined with weather information was analyzed to better understand the factors that may contribute to flight delays. Since the original dataset contained a very large number of records, several preprocessing steps were performed, including data cleaning and sampling, before starting the analysis.

Different data analysis and visualization techniques were used to explore the dataset and understand its characteristics. These included correlation analysis, histograms, box plots, and scatter plots. The visualizations helped provide a clearer picture of the data and made it easier to identify patterns and relationships between different variables.

The analysis showed that some weather-related factors, such as wind speed, may have an impact on flight delays. However, the results also indicated that delays cannot be explained by a single factor alone and are likely influenced by a combination of weather and operational conditions.

A Decision Tree model was also developed to predict flight delays using selected weather and operational features. The model achieved an accuracy of 69.4%, which demonstrates that machine learning methods can be useful for studying and predicting flight delays.

Overall, this project highlighted the importance of data analysis and visualization before building predictive models. It also showed that combining weather data with operational flight information can provide a better understanding of the causes of flight delays and support decision-making in the aviation industry.